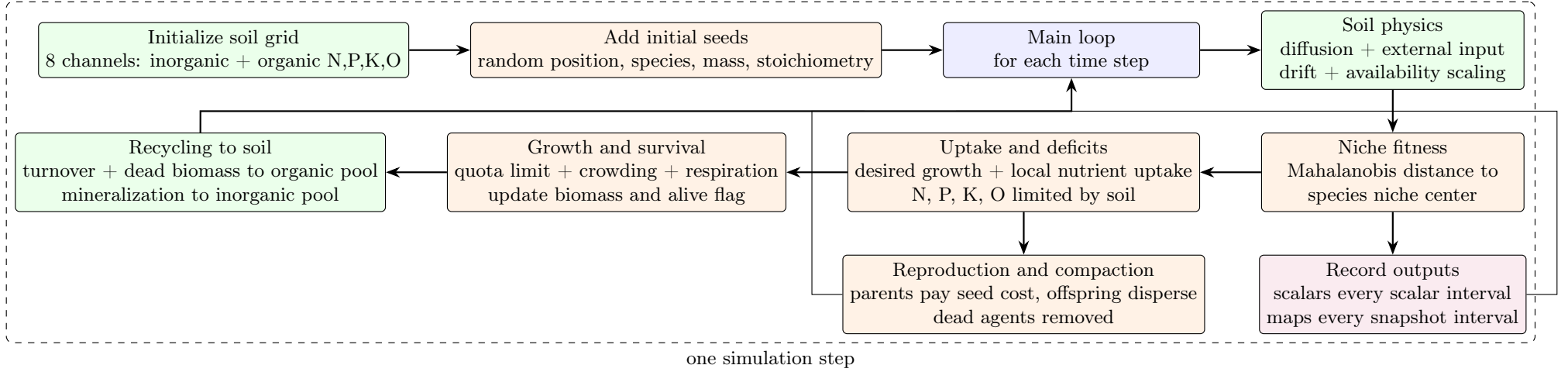


Hybrid Ecosystem Simulator – Simulation Flow Schematic



Flow summary. The model initializes a spatial soil tensor and an agent buffer, inserts species-specific seeds, and then iterates a time-step loop. In each step it updates soil physics, computes stoichiometric fitness, calculates nutrient uptake and deficits, applies growth and mortality, recycles biomass into soil, handles reproduction, and records scalar or spatial outputs when the configured intervals are reached.

- Soil stores inorganic and organic N, P, K, and O pools.
- Each agent stores position, species identity, biomass, elemental composition, and alive state.
- Fitness is based on Mahalanobis distance between current elemental composition and the species niche center.
- Growth is limited by nutrient availability, internal quotas, respiration, and local crowding.
- Outputs include biomass, agent counts, element pools, per-species summaries, nutrient deficits and spatial biomass maps.